

SVD (Singular Value Decomposition)

A matrix $A \in \mathbb{C}^{m \times n}$ permits the following factorization: -

$$\underbrace{A}_{m \times n} = \underbrace{U}_{m \times m} \underbrace{\begin{bmatrix} \overbrace{\sigma_1 \dots \sigma_r}^{r \times r} & \overbrace{0}^{r \times (n-r)} \\ \underbrace{0}_{(m-r) \times r} & \underbrace{0}_{(m-r) \times (n-r)} \end{bmatrix}}_{m \times n} \underbrace{V^H}_{n \times n}$$

$$U U^H = I_{m \times m}, \quad \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$$
$$V V^H = I_{n \times n}.$$

$\sigma_1, \dots, \sigma_r$ are 'singular values' of A .

Properties implied by SVD: (All of these properties can be proved)

Notation: $\Sigma = \begin{bmatrix} \sigma_1 & \dots & \sigma_r & 0 \\ 0 & & & 0 \end{bmatrix}$

① Rank(A) = r

SVD is rank-revealing decomposition

② $A^H A = V \Lambda V^H$, $(A = U \Sigma V^H)$

where $\Lambda = \Sigma^H \Sigma$

Columns of V are eigenvectors of

$A^H A$,

What are eigenvalues of $A^H A$?

$$\underbrace{\Sigma^H \Sigma}_{n \times n} = \begin{bmatrix} \sigma_1^2 & & & 0 \\ & \sigma_2^2 & & \\ & & \ddots & \\ & & & \sigma_r^2 & \\ 0 & & & & 0 \end{bmatrix}$$

They are $\sigma_1^2, \sigma_2^2, \dots, \sigma_r^2, 0$ (can be repeated)

③ $AA^H = U \underbrace{\Sigma \Sigma^H}_{\Lambda} U^H$

The columns of V are eigenvectors of AA^H

The eigenvalues of AA^H are $\sigma_1^2, \sigma_2^2, \dots, \sigma_r^2, 0$ (possibly repeated)

4) Fundamental subspaces associated with A :

$$A = U \begin{bmatrix} \sigma_1 & \dots & \sigma_r & 0 \\ 0 & & 0 & \end{bmatrix} V^H$$

$$= \underbrace{\begin{bmatrix} U_1 & U_2 \end{bmatrix}}_{\substack{m \times r \quad m \times (m-r)}} \begin{bmatrix} \sigma_1 & \dots & \sigma_r & 0 \\ 0 & & 0 & \end{bmatrix} \underbrace{\begin{bmatrix} V_1^H \\ V_2^H \end{bmatrix}}_{\substack{n \\ r \quad n-r}}$$

Basis for $R(A)$:

The columns of U_1 form a basis for $R(A)$.

Proof:

$$A = U \Sigma V^H$$

$$AV = U \Sigma = [U_1, U_2] \begin{bmatrix} \sigma_1 & & 0 \\ & \ddots & \\ 0 & \sigma_r & 0 \\ & & \ddots & \\ 0 & & 0 & \end{bmatrix}$$

$$A [U_1, U_2] = [U_1, \begin{bmatrix} \sigma_1 & & 0 \\ & \ddots & \\ 0 & \sigma_r & 0 \end{bmatrix}, 0]$$

$$\Rightarrow AV_1 = U_1 \underbrace{\begin{bmatrix} \sigma_1 & & 0 \\ & \ddots & \\ 0 & \sigma_r & 0 \end{bmatrix}}_{\Sigma_r}, AV_2 = 0$$

$$\Rightarrow U_1 = AV_1 \Sigma_r^{-1}$$

$\Rightarrow$ every column of U_1 belongs to

$$R(A).$$

$$\Rightarrow R(U_1) \subseteq R(A)$$

$$\text{But } \dim R(U_1) = r$$

$$\Rightarrow \boxed{\begin{array}{l} \dim R(A) = r \\ R(U_1) = R(A) \end{array}}$$

In particular, columns of U_1 form an orthonormal basis for $R(A)$.

Basis for $N(A)$:

Justify that columns of U_2 form an orthonormal basis for $N(A)$.

Orthonormal Basis of a Subspace:

A Closer Look

Let S be a subspace of $\mathbb{C}^n$ of dimension k
and let $v_1, v_2, \dots, v_k$ be an orthonormal basis of S .

Define: $U = [v_1 \dots v_k]$

$$U^H U = I_{k \times k}$$

$$U U^H \neq I$$

$$k < n.$$

Consider any vector $y \in \mathbb{C}^n$
We wish to compute its orthogonal projection onto S .

Normal equations:

$$\left[\hat{y}_S = U x \right] \underbrace{U^H U}_{I_{k \times k}} = U^H y$$

$$x = U^H y$$

$$\hat{y}_S = Ux = (UU^H)y$$

The above is true for any vector

y

$$\hat{y}_S = UU^H y$$

$$= P_S y \quad (P_S := UU^H)$$

UU^H : orthogonal projection matrix

$$\text{rank}(UU^H) = k \quad (\text{verify})$$

What is projection matrix onto

$N(A)$?

$$V_2 V_2^H = P_{N(A)}$$

EXISTENCE OF SVD

Consider the matrix

$$A \in \mathbb{C}^{m \times n}$$

$$A^H A \in \mathbb{C}^{n \times n}$$

(i) $A^H A$ is Hermitian symmetric

$$(A^H A)^H = A^H (A^H)^H = A^H A$$

(ii) $A^H A$ is also PSD

$$x^H \underbrace{A^H A}_{\substack{= \\ U^H U}} x = u^H u = \|u\|_2^2 \geq 0$$

$\Rightarrow A^H A$ can be diagonalized by a unitary matrix

$$\Rightarrow A^H A = V \Lambda V^H, \text{ for some unitary } V$$

$$\text{and } \Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix},$$

$$\lambda_i \geq 0, \quad i=1, 2, \dots, n$$

$$\text{If } \text{rank}(A) = r$$

What is $\text{rank}(A^H A)$? It is "r"

(DIY)

$$\dim(N(A^H A)) = n - r$$

$A^H A$ has " $n - r$ " zero eigenvalues

$$\Rightarrow \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r > 0$$

$$\text{and } \lambda_{r+1} = \lambda_{r+2} = \dots = \lambda_n = 0$$

$$\Rightarrow A^H A = V \begin{bmatrix} \lambda_1 & & & \\ & \ddots & & \\ & & \lambda_r & \\ & & & 0 & \ddots & 0 \end{bmatrix} V^H$$

Define

$$\sigma_i = \sqrt{\lambda_i}, \quad i = 1, 2, \dots, r$$

$$\sigma_i > 0$$

$$\Rightarrow \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r \geq 0.$$

Define:

for $i=1, 2, \dots, r$

$$u_i := \frac{1}{\sigma_i} A v_i$$

Then,

$$u_i^H u_i = \frac{1}{\sigma_i^2} v_i^H A^H A v_i$$

$$= \frac{\lambda_i}{\sigma_i^2} v_i^H v_i = 1$$

$i=1, 2, \dots, r$

Similarly -

$$u_i^H u_j = \frac{1}{\sigma_i \sigma_j} v_i^H \underbrace{A^H A v_j}_{\lambda_j v_j}$$

$$U_1 = \frac{1}{\sigma_1} A v_1$$

$$U_2 = \frac{1}{\sigma_2} A v_2$$

⋮

$$U_r = \frac{1}{\sigma_r} A v_r$$

$$\begin{bmatrix} U_1 & \dots & U_r \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix} = A \begin{bmatrix} v_1 & \dots & v_r \end{bmatrix}$$

$$A \begin{bmatrix} v_1 & \dots & v_r \end{bmatrix}$$

$$= \begin{bmatrix} U_1 & \dots & U_r & U_{r+1} & \dots & U_m \end{bmatrix} \begin{bmatrix} \sigma_1 & & & & \\ & \ddots & & & \\ & & \sigma_r & & \\ & & & \mathbf{0} & \end{bmatrix}$$

What is

$$A v_{r+1} = 0$$

$$A v_{r+2} = 0$$

$$\vdots$$

$$A v_n = 0 \quad \checkmark$$

$$\Rightarrow A \begin{bmatrix} v_1 & \dots & v_r & v_{r+1} & \dots & v_n \end{bmatrix}$$

$$= \underbrace{\begin{bmatrix} v_1 & \dots & v_r & v_{r+1} & \dots & v_n \end{bmatrix}}_V \begin{bmatrix} \lambda_1 & \vdots & 0 \\ \vdots & \lambda_r & \vdots & 0 \\ \vdots & \vdots & \ddots & \vdots & 0 \\ 0 & \vdots & \vdots & \vdots & 0 \end{bmatrix}$$

Σ

$$AV = U\Sigma$$

$$\Rightarrow A = U\Sigma V^H$$

SPECTRAL NORM OF

A MATRIX

consider the ratio

$$\max_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2}$$

(A need not be symmetric)

Spectral Norm of A ,
denoted by $\|A\|_2$ is
defined as

$$\|A\|_2 = \max_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2}$$

Note:

$$\frac{\|Ax\|_2^2}{\|x\|_2^2} = \frac{x^H A^H A x}{x^H x} \\ = R_{A^H A}(x)$$

$$\max_{x \neq 0} R_{A^H A}(x) = \max \text{ eigenvalue of } A^H A$$

$$= \sigma_1^2$$

$$\Rightarrow \|A\|_2 = \sigma_1$$

$$\Rightarrow \|Ax\|_2 \leq \sigma_1 \|x\|_2$$